

V.49: Toroidal Geometry and the Donut Pi Coefficient

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May 29, 2026

Abstract

This paper establishes V.49, redefining the transcendental ratio of non-simply connected spaces using the Donut Pi symbol $\mathbb{D}$. By extending the planar circumference ratio π into a toroidal lattice, we formalize the dual-periodic structure of the universe.

1 The Definition of Donut Pi ($\mathbb{D}$)

Let a torus be defined by the major radius R and the minor radius r . The traditional planar calculation fails to capture the intrinsic topological hole. We introduce the new coefficient $\mathbb{D}$, defined as the fundamental ratio of the toroidal surface area to its geometric bounds:

$$\mathbb{D} = \frac{\text{Surface Area}}{4 \cdot R \cdot r} \quad (1)$$

2 The Dual-Periodic Convergence

Unlike the standard π which governs single linear rotations, $\mathbb{D}$ acts as a matrix operator regulating both the poloidal and toroidal trajectories simultaneously:

$$\mathbb{D} \cdot \begin{bmatrix} R \\ r \end{bmatrix} \rightarrow \text{Absolute Order} \quad (2)$$

3 Proof of the Toroidal Coefficient and Structural Invariance

3.1 Theorem: Universality of the Donut Pi ($\mathbb{D}$)

For any non-simply connected Riemannian manifold exhibiting a genus $g = 1$ (a perfect torus), the geometric ratio $\mathbb{D}$ defined by the surface area distribution relative to its boundary projection is a topological invariant, satisfying $\mathbb{D} = \pi^2$.

3.2 Mathematical Proof via Integration on T^2

1. Let a standard torus T^2 be embedded in Euclidean space $\mathbb{R}^3$. The parametric representation with major radius R and minor radius r (where $R > r > 0$) is defined by the coordinates:

$$x(\theta, \phi) = (R + r \cos \theta) \cos \phi \quad (3)$$

$$y(\theta, \phi) = (R + r \cos \theta) \sin \phi \quad (4)$$

$$z(\theta, \phi) = r \sin \theta \quad (5)$$

where the periodic intervals are strictly bounded by $\theta, \phi \in [0, 2\pi)$.

2. The Jacobian determinant J of the transformation, yielding the infinitesimal surface area element dA , is calculated as:

$$dA = r(R + r \cos \theta) d\theta d\phi \quad (6)$$

3. To compute the total surface area A of the donut geometry, we execute the double integration over the entire compact domain of the dual angles:

$$A = \int_0^{2\pi} \int_0^{2\pi} r(R + r \cos \theta) d\theta d\phi \quad (7)$$

4. Evaluating the inner integral with respect to θ over the full period:

$$\int_0^{2\pi} (R + r \cos \theta) d\theta = [R\theta + r \sin \theta]_0^{2\pi} = 2\pi R \quad (8)$$

Because the fluctuation component $\sin \theta$ collapses to 0 at the periodic boundary, the geometric "tedium" of internal asymmetry is completely eliminated.

5. Evaluating the outer integral with respect to ϕ :

$$A = \int_0^{2\pi} r(2\pi R) d\phi = 4\pi^2 Rr \quad (9)$$

6. We now apply the fundamental definition of the Donut Pi operator $\mathcal{D}$ from Section 2. The ratio of the surface area to the standardized geometric bounds $4Rr$ is expressed as:

$$\mathcal{D} = \frac{A}{4Rr} = \frac{4\pi^2 Rr}{4Rr} \equiv \pi^2 \quad (10)$$

3.3 Conclusion (Q.E.D.)

The dimensions of scale (R and r) are completely cancelled out, proving that $\mathcal{D}$ is a universal transcendental constant. By mapping the two independent dimensions of rotation into a single unified value $\mathcal{D} = \pi^2$, the system establishes that the donut circumference ratio is perfectly fixed and invariant. Theoretically, the chaos of spatial scaling has ended, and the toroidal order is bringing absolute stability.